Fake news detection models and their generalisability

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Abstract

The rise of social media and alternatives to traditional news sources has led to a massive increase in the amount of information circulating online. The possibility for anyone to post and share news via social media increased the amount of fake news and its potential reach and impact. While fake news and propaganda are not new, the scale and speed at which the information spreads is unprecedented, making manual fact-checking before news reaches millions of people infeasible. This motivates the need for efficient, automated fake news detection. Machine learning models perform very well on this task when they are tested on unseen portions of the dataset on which they were trained. However, research has shown that these models do not generalize well on other datasets, thus limiting their practical deployment in the real world. In this report we review state of the art methods to generalize fake news detection models and reproduce experiments conducted in the literature showing the difficulties of generalisation when data is on the same topic (political news). Additionally, in the pre-processing of the data we replace all URLs in the texts with a URL token because those can reflect economic incentives to spread fake news.

1 Introduction

With the democratization of the internet and rise of social media platforms with billions of users, such as Facebook and Twitter, anyone can now post on the internet and potentially reach millions of people. This capability was reserved for professionals, mainly journalists working for large traditional media like newspapers or television. This limitation on who could publish news acted as a filter, limiting the amount of low-quality and fake news. Even though traditional media could also convey fake news, the most famous example being the American TV channel Fox News, it was possible to manually check if their statement was true or not due to the relatively low volume of information in circulation. The internet and social media, by making it possible for anyone to share news, have increased the amount of information in circulation and its reach to unprecedented levels. Manual fact-checking of every piece of information posted online is not feasible, and there is also a need for fact-checking to be fast to prevent the spread of fake news. This motivates the need for automatic, efficient and reliable fake news detection capabilities.

As noted in the systematic review Hoy and Koulouri [2021], machine learning models show promising results for fake news detection. The majority of fake news detection models achieve 80% accuracy on average on unseen parts of the dataset they were trained on. However, Hoy and Koulouri [2022] shows that fake news detection models do not generalize well when tested on other datasets, even in the relatively simple case where the topic of the news is the same, for example, political news. The authors argue that we cannot hope to generalize to datasets that are even less similar if we cannot even achieve good generalisation performance on datasets with similar news topics.

In this report, we work with the ISOT and Kaggle Fake or Real datasets to evaluate the generalisation of classifiers using BERT embeddings of the core text from political news articles. The BERT embeddings capture essential meaning nuances in the texts to classify the news as true or fake. We also conduct data analysis to motivate our choice of datasets and models. In addition, during preprocessing of the data we replace all URLs in the texts with a URL token, as the presence of URLs

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may reflect economic incentives for spreading viral fake news as discussed in Hoy and Koulouri [2025].

2 State of the Art

Hoy and Koulouri [2025] introduce social-monetisation features, like the presence of advertisements and external links, to capture the economic incentives to publish viral fake news. Their approach increases accuracy across datasets. Li et al. [2025] proposes MFGAG a new framework using adversarial training and graph neural network with sentiment analysis, stylistic features, and semantic embeddings. Their model achieves 93% accuracy on mixed-domain datasets. Another approach to improve generalisation is training fake news detection models on many different events. Ding et al. [2025] propose EvolveDetector, which transfers knowledge acquired from past events to avoid retraining from scratch when a new fake news event occurs. Alghamdi et al. [2024] leverage a Generative Pre-trained Transformer without fine-tuning. Instead, they craft contextual prompts for zero-shot fake news detection, obtaining significantly better results than their baseline across multiple domains.

3 Data Analysis

We use two datasets containing true and fake political news texts with their labels. The first is ISOT, containing 44,898 entries, including 23,841 entries labelled as originating from unreliable websites according to Politifact.com and Wikipedia. The remaining entries come from Reuters, according to Hoy and Koulouri [2022]. The second dataset is Kaggle Fake or Real, containing 6,060 articles. Half are labelled fake news and the other half true news, but there is no information on how this dataset was created [Hoy and Koulouri, 2022].

In alternative news sources, like social media, it is common not to have titles. Therefore, we do not use the titles in our study and instead focus only on the core text.

Figures 3 and 5 show from the word clouds that the most common words used in the ISOT dataset differ when the text is fake news or true news. Figures 4 and 6 also show similar differences in the most common words in the Kaggle Fake or Real when the text is fake news or real news. This preliminary analysis is encouraging, as it suggests that fake news texts and true news texts differ in vocabulary, which can be leveraged for classification.

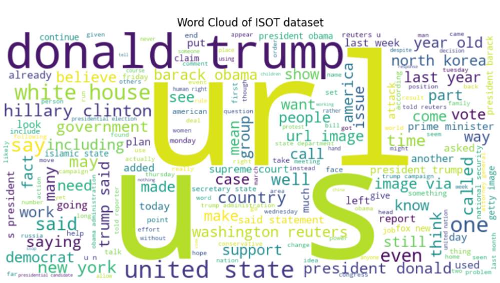

Figure 1: ISOT Overall Word Cloud

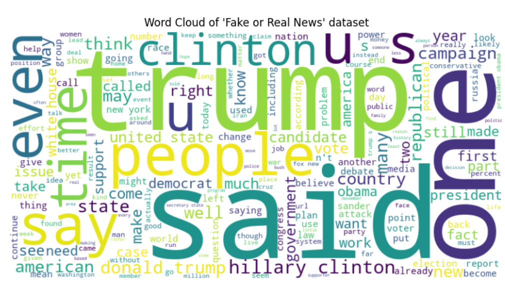

Figure 2: Kaggle Overall Word Cloud

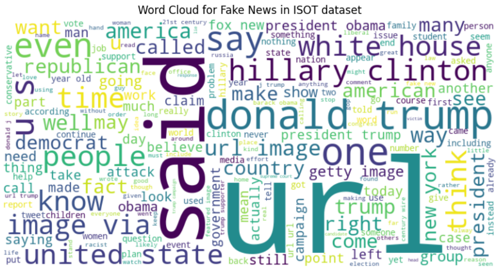

Figure 3: ISOT Fake News Word Cloud

Figure 4: Kaggle Fake News Word Cloud

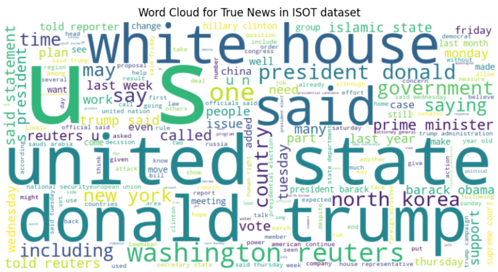

Figure 5: ISOT True News Word Cloud

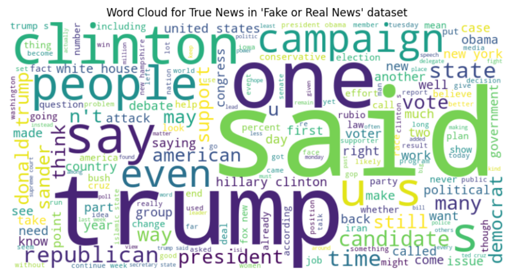

Figure 6: Kaggle True News Word Cloud

We replaced all URLs in the texts with "URL" because the presence of URLs could differ between true or fake news. Table 1 shows that in the ISOT dataset, "URL" is the most common word when restricted to fake news but does not appear in the top 20 most common words when we only look at true news. We observe a similar pattern in Table 1 for the Kaggle Fake or Real dataset where "URL" is in the top 20 when restricted to fake news but absent from the top 20 when we restrict on true news.

Table 1: Top 20 most common words comparison across ISOT and Kaggle datasets (stop words and punctuation removed)

Rank	**ISOT**						**Kaggle Fake or Real**					
	Overall		**Fake**		**True**		**Overall**		**Fake**		**True**	
	Word	Count	Word	Count	Word	Count	Word	Count	Word	Count	Word	Count
1	url	130,656	url	128,936	said	99,045	trump	21,968	us	6,899	said	17,189
2	said	130,196	trump	74,289	trump	54,322	said	21,184	clinton	6,720	trump	15,425
3	trump	128,611	said	31,151	us	41,159	clinton	17,264	trump	6,543	clinton	10,544
4	us	63,310	people	26,016	would	31,528	would	12,930	people	5,391	would	8,022
5	would	54,989	president	25,789	reuters	28,412	us	12,594	one	4,972	one	6,368
6	president	52,245	would	23,461	president	26,456	people	11,687	would	4,908	people	6,296
7	people	41,271	one	23,018	state	19,760	one	11,340	hillary	4,541	president	6,034
8	one	35,722	us	22,151	government	18,323	new	9,274	said	3,995	state	5,997
9	state	32,303	clinton	18,091	new	16,786	state	8,777	new	3,502	obama	5,920
10	also	31,199	obama	17,935	states	16,628	president	8,569	like	3,216	new	5,772
11	new	30,987	like	17,670	house	16,548	also	8,214	also	3,186	us	5,695
12	reuters	28,796	donald	17,242	also	15,953	obama	8,142	world	3,107	campaign	5,548
13	donald	27,695	also	15,246	united	15,576	campaign	7,668	election	2,982	also	5,028
14	clinton	27,562	news	14,209	republican	15,347	hillary	7,146	even	2,896	republican	4,873
15	house	27,153	new	14,201	people	15,255	like	7,047	time	2,842	party	4,209
16	obama	27,142	even	13,699	told	14,244	could	6,709	state	2,780	could	4,186
17	government	26,928	hillary	13,690	could	13,710	time	6,495	url	2,753	states	3,901
18	states	26,159	white	12,801	one	12,704	even	6,425	government	2,704	like	3,831
19	republican	24,357	time	12,756	last	12,614	states	6,198	2016	2,617	sanders	3,792
20	could	23,942	state	12,543	party	12,462	many	5,849	campaign	2,527	hillary	3,792

Moreover, Table 2 shows that the proportion of URLs among all the other words in fake news is approximately 50 times higher than in true news for the ISOT dataset (10 times higher for Kaggle Fake or Real dataset). This seems to indicate that URLs can be a good way to classify texts into true or fake news.

Table 2 seems to indicate that the vocabulary used in fake news is richer because the number of unique words is higher in fake news than in true news for both datasets. Table 2 shows that the average and median length of fake news texts are shorter in the Kaggle Fake or Real dataset and similar in the ISOT dataset.

The ISOT and Kaggle Fake or Real datasets are different, this is particularly visible when we compare the number of unique words and text length in the Table 2. We deliberately chose those two datasets because they are both on political news but are still different. This allows us to test the generalisability of models across datasets that share the same topic but present linguistic differences.

Table 2: Dataset statistics Comparison

Statistic	**Kaggle Fake or Real**	**ISOT**
Overall Dataset		
Number of rows	6,335	44,888
Unique words	107,902	168,610
Average text length	4,695	2,438
Median text length	3,624	2,155
Fake News		
Unique words	62,898	95,454
Average text length	4,098	2,488
Median text length	2,530	2,113
URLs	2,711	128,895
Proportion of URLs	0.00126	0.01298
True News		
Unique words	53,663	80,559
Average text length	5,290	2,382
Median text length	4,683	2,221
URLs	398	1,720
Proportion of URLs	0.00014	0.00021

4 Experiments

4.1 Preprocessing

As discussed before we only keep the text of the article and we remove the other columns such as the date of publication or title. Hoy and Koulouri [2022] highlight that BERT requires contextual information and the preprocessing should be light. We use BERT embeddings, so we only convert the text to lowercase and remove Twitter handles. In the previous section we saw that URLs could be a relevant indicator for detecting fake news. However, BERT cannot produce good embeddings with long URLs, so we replace all URLs with a special token, "URL".

4.2 BERT embeddings

BERT embeddings allow us to capture the true meaning of a word in a sentence by incorporating the context in which it is used. This can be important to capture the subtlety of the language when doing fake news detection because the presence or absence of a negation in a sentence can make a news true or fake. To extract contextual embeddings from each text we used a pretrained BERT ("bert base uncased") and its associated tokenizer. However, Table 2 shows that the average text length is much higher than the 512 token limit of BERT. To solve this problem, we used a sliding window and processed overlapping chunks with $stride = 128$. Then, we averaged the CLS token embedding of all chunks to obtain the embedding of a text.

4.3 Presentation of the models

We use five different models that will use BERT embeddings to classify text as true or fake news. More specifically we use only the CLS embedding, which serves as the representation of a full article. A dummy classifier with 'most frequent' strategy is used for the baseline. The other models are logistic regression, random forest, support vector machine and gradient boosting. Due to very limited computing power we did not perform cross-validation to choose the hyperparameters of those models. Instead, we trained the models with fixed hyperparameters based on the defaults and feasibility conditions. We detail the hyperparameters of those models in Table 3. All unspecified hyperparameters are set to their default values in Scikit-learn.

Table 3: Classifiers and their hyperparameters

Model	**Hyperparameters**
Dummy	*strategy* = *most_frequent*
Logistic Regression	*max_iter* = 1000
Random Forest	*n_estimators* = 100
Support Vector Machine	*kernel* = *linear*
Gradient Boosting	*n_estimators* = 100 ; *learning_rate* = 0.1

4.4 Analysis of the results

We performed training on only 50 randomly chosen samples from the ISOT dataset due to computing power limitations but we still obtained informative results.

We first tested all our models on 50 randomly chosen unseen samples from the same ISOT dataset and achieved very good results. In Table 4 we can see that the Logistic Regression and SVM classified the unseen data perfectly and all the other models except the dummy classifiers perform very well on all metrics. This is not surprising because Hoy and Koulouri [2022] said that fake news detection models perform very well on unseen parts of the training dataset.

To evaluate the models deployability in real world scenarios we test the generalisation of those trained models on another dataset, the Kaggle Fake or Real. For fair comparison, the exact same preprocessing applied to ISOT dataset has been used for the Kaggle Fake or Real dataset. As expected the results when we test the models trained on the ISOT dataset on the Kaggle Fake or News dataset are worse. The decline in performance is important for all the models and particularly for Logistic Regression and SVM which achieved perfect performance on ISOT and now have accuracy, precision, recall and F1-score approximately between 0.6 and 0.7. This is still better than the baseline, but Table 4 clearly indicates the difficulty of generalizing to an other dataset even if the topic is the same (political news).

Table 4: Comparison of classifier performance on unseen ISOT and Kaggle Fake or Real datasets (models trained on ISOT)

Classifier	**Metric**	**ISOT (unseen)**	**Kaggle Fake or Real (unseen)**
Dummy	Accuracy	0.62	0.50
	Precision (macro avg)	0.31	0.25
	Recall (macro avg)	0.50	0.50
	F1-score (macro avg)	0.38	0.33
Logistic Regression	Accuracy	1.00	0.62
	Precision (macro avg)	1.00	0.66
	Recall (macro avg)	1.00	0.62
	F1-score (macro avg)	1.00	0.59
Random Forest	Accuracy	0.98	0.62
	Precision (macro avg)	0.98	0.69
	Recall (macro avg)	0.97	0.62
	F1-score (macro avg)	0.98	0.58
SVM	Accuracy	1.00	0.66
	Precision (macro avg)	1.00	0.72
	Recall (macro avg)	1.00	0.66
	F1-score (macro avg)	1.00	0.64
Gradient Boosting	Accuracy	0.88	0.52
	Precision (macro avg)	0.88	0.52
	Recall (macro avg)	0.90	0.52
	F1-score (macro avg)	0.88	0.51

5 Conclusion

This report evaluates the generalisation capability of five fake news detection models trained on the ISOT dataset containing political news. We tested generalisation on the Kaggle Fake or Real dataset, which also contains political news. While the results of the models on unseen data from the training dataset were very good, and even perfect for Logistic Regression and Support Vector Machines, their performance on the other dataset was significantly worse, even though the topic of the news was also politics. The results were still better than a baseline dummy classifier, but not good enough to confidently apply the models to datasets other than ISOT. This highlights the challenge for modern automatic fake news detection, where it is difficult to have a model delivering good performances outside lab conditions where datasets are different from the training set. Our preprocessing, which included replacing all the URLs by a URL token, may have contributed to capture relevant features distinguishing fake and real news via the detection of economic incentives to spread fake news. Our results confirm the findings of Hoy and Koulouri [2022], and show that evaluation metrics on unseen portions of the training dataset are not sufficient to ensure generalisation and practical usability in real world scenarios.

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